

MVA Lab

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June 9, 2026

Abstract

The aim of this lab exercise is the selection of $B_s^0 \rightarrow \psi(2S)K_s^0$ rare decay events from a large combinatorial background component in LHCb experiment data. A multivariate classification approach is utilized to isolate the signal from the background. Utilizing Monte Carlo simulations for the signal and the kinematically similar $B^0 \rightarrow \psi(2S)K_s^0$ decay as a control channel. Discriminatory features are selected and used to train the classifier. The methodology demonstrates the effectiveness of machine learning algorithms in high-energy particle physics data analysis for extracting rare signal events while ensuring variables remain uncorrelated to the invariant mass to avoid biased fits. We record a signal with approximate significance of 4.45 at the end.

1 Introduction

Modern particle physics relies on probing the fundamental building blocks of the universe through high-energy particle collisions. The Large Hadron Collider (LHC) at CERN is at the forefront of this endeavor, accelerating and colliding proton bunches at a centre of mass energy of $\sqrt{s} = 13$ TeV. During its Run 2 operational period (2015–2018), these collisions occurred at a rate of 40×10^6 times per second. The LHCb (Large Hadron Collider beauty) experiment, one of the four main detectors situated at the LHC interaction points, is specifically optimized for the study of heavy-flavour physics, predominantly hadrons containing a bottom (b) or charm (c) quark. Precision measurements of these decays allow physicists to test the Standard Model and search for hints of New Physics.

Following a proton-proton collision, the constituent partons undergo hadronization, producing various particles. Among these, b -hadrons, such as the B_s^0 meson (composed of a strange quark s and a bottom antiquark $\bar{b}$), are the one in which LHCb is interested. However, these heavy hadrons are unstable. They usually propagate less than a centimetre from the primary interaction vertex before decaying, long before reaching the active tracking components of the detector. Consequently, their presence and properties must be inferred by fully reconstructing their kinematic profiles from their lighter, long-lived decay products. In this study, we focus on the specific rare decay channel $B_s^0 \rightarrow \psi(2S)K_s^0$ shown in figure 1.

Identifying this rare signal is a challenge. The high luminosity of the LHC environment produces an high flux of particles, resulting in a large combinatorial background: random combinations of uncorrelated particle tracks that accidentally mimic the kinematics of the true signal decay. Because the fraction of interesting signal events is small if compared to the overwhelming background, simple, one-dimensional selection criteria (cut-based

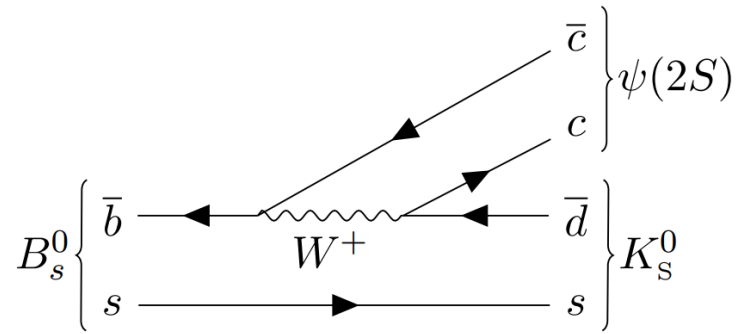


Figure 1: *Leading order Feynman diagram of the decay $B_s^0 \rightarrow \psi(2S) K_S^0$*

analysis) are insufficient to isolate the signal without sacrificing a significant portion of true signal candidates.

To address this, modern high-energy physics analyses heavily rely on Machine Learning (ML) techniques. This report details the implementation of a MultiVariate Analysis (MVA) to optimally separate the $B_s^0 \rightarrow \psi(2S) K_S^0$ signal from the heavily polluted data. By evaluating multiple kinematic and geometric features simultaneously, multivariate classifiers can exploit subtle correlations between variables to maximize the signal-to-background ratio and obtain the best possible performance.

Crucially, the classification algorithm must be trained and validated. We utilize a combination of real LHCb data (defining the background from the upper invariant mass sideband) and Monte Carlo (MC) simulations (for the signal). Furthermore, to account for potential mismodelings of decay kinematics and detector responses in the simulation, a kinematically similar control channel, $B^0 \rightarrow \psi(2S) K_S^0$, is utilized to validate the features. A fundamental constraint applied during this feature selection is that the variables used for ML training must remain largely uncorrelated with the invariant mass of the B_s^0 candidate. This strict requirement ensures that the ML selection does not artificially create a false peak in the background mass distribution, allowing for unbiased extraction of the final signal.

1.1 The LHCb Detector

Unlike the general-purpose detectors at the LHC, which are built with a cylindrical geometry to capture particles in all directions, the LHCb (Large Hadron Collider beauty) experiment is designed as a single-arm forward spectrometer. It covers a pseudorapidity range of $2 < \eta < 5$. This specific geometry is driven by the kinematics of heavy-quark production at high energies: at the LHC, the b and $\bar{b}$ quarks are predominantly produced in the same forward (or backward) cone, highly boosted along the beam axis.

To successfully reconstruct rare decays such as $B_s^0 \rightarrow \psi(2S) K_S^0$, the detector relies on several highly specialized sub-systems designed for precise tracking, vertexing, and particle identification (PID):

- **Vertex Locator (VELO):** Positioned few millimeters from the proton beam, the VELO is a silicon microstrip detector that precisely reconstructs the primary proton-proton interaction vertex as well as the displaced secondary vertices of long-lived particles. Because b -hadrons typically fly a few millimeters before decaying, the

VELO's spatial resolution is the most critical tool for distinguishing signal decay vertices from the prompt combinatorial background originating at the primary vertex.

- **Tracking System and Dipole Magnet:** The tracking system consists of stations located both upstream and downstream of a dipole magnet (providing a bending power of 4 Tm). By measuring the deflection of charged particles, the tracking system determines their momentum with high precision. This is essential for accurately calculating the invariant mass of the B_s^0 candidate from its final-state products.
- **Particle Identification (RICH and Muon Stations):** Robust PID is required to separate the signal from vast background processes. Two Ring Imaging Cherenkov (RICH) detectors provide excellent separation between pions, kaons, and protons, which is important for correctly identifying the pion tracks from the $K_S^0 \rightarrow \pi^+\pi^-$ decay. Finally, the furthest sub-systems are the Muon Chambers, composed of alternating layers of iron and multi-wire proportional chambers. These stations reliably identify the μ^+ and μ^- tracks originating from the $\psi(2S)$ decay, providing a clean signature to trigger on and record the event.

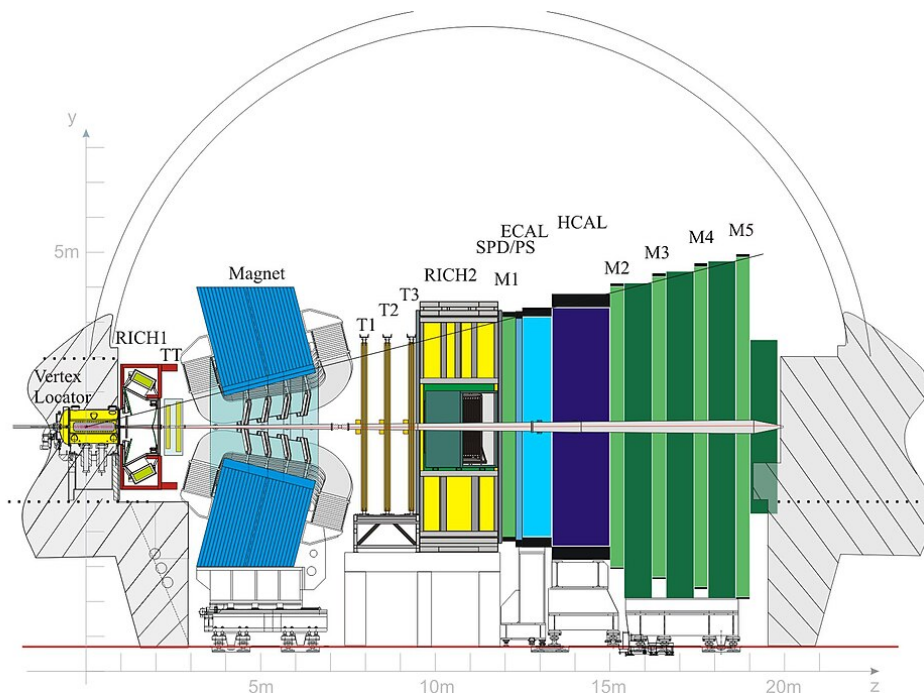


Figure 2: Schematic side view of the LHCb detector, showing the forward-arm geometry and the arrangement of the various sub-detectors along the beam axis. The interaction point is located on the far left inside the VELO.

Through the combined performance of these sub-systems, the LHCb detector provides the necessary mass resolution, decay-time resolution, and particle identification capabilities required to isolate the $B_s^0 \rightarrow \psi(2S)K_S^0$ signal for multivariate analysis.

2 Data Analysis

To isolate the $B_s^0 \rightarrow \psi(2S)K_S^0$ signal, this analysis relies on three distinct datasets:

- **Real Data:** The dataset recorded by the LHCb experiment after preliminary reconstruction and rough pre-selection criteria. See figure 3 for have a look at the plot of the invariant mass.
- **Signal Simulation:** A Monte Carlo (MC) simulation sample of the target signal decay $B_s^0 \rightarrow \psi(2S)K_S^0$. See figure 4 to observe the distribution of the simulated invariant mass.
- **Control Simulation:** A MC simulation sample of the kinematically similar decay $B^0 \rightarrow \psi(2S)K_S^0$.

In typical high-energy physics analyses, the rare decay of interest is initially hidden in massive amounts of background. In our real dataset, the most prominent peak visible in the invariant mass distribution is not our signal, but rather a large peak corresponding to the $B^0 \rightarrow \psi(2S)K_S^0$ decay. Because this decay shares the exact same final state particles and possesses extremely similar kinematics to our signal, these events easily pass the initial selection requirements.

For the purpose of this analysis, we assume the data sample is composed of three components:

1. **Combinatorial Background:** The dominating background component, consisting of random tracks that happen to form a vertex and mimic the signal kinematics.
2. **The B^0 Peak:** The abundant, visible control channel.
3. **The B_s^0 Signal:** The rare events we aim to extract.

To study the physics of the $B_s^0 \rightarrow \psi(2S)K_S^0$ decay, we must improve the signal-to-background ratio. The objective of this exercise is to train a machine learning classifier to distinguish between the signal and the combinatorial background.

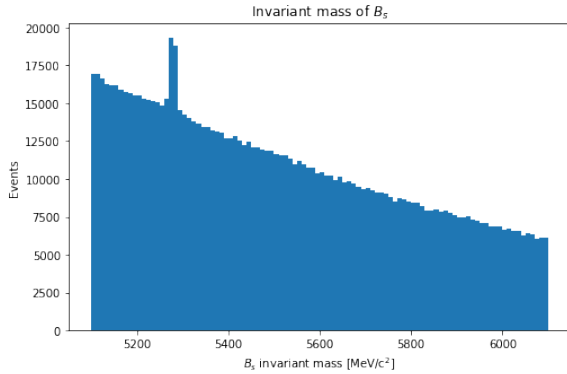


Figure 3: Invariant mass distribution of the reconstructed child particles in the full LHCb dataset. The visible peak is dominated by the $B^0 \rightarrow \psi(2S)K_S^0$ control decay, embedded in a large combinatorial background.

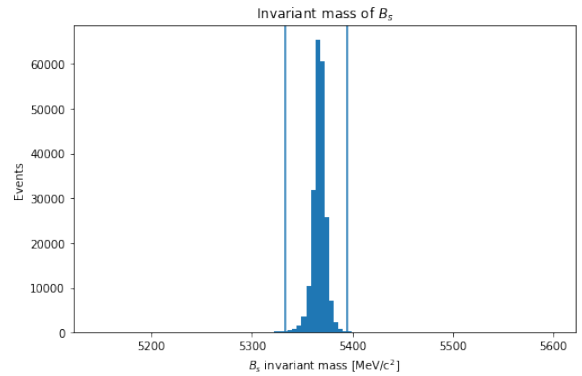


Figure 4: Simulated invariant mass distributions for the B_s^0 signal (target). The two vertical lines individuate the signal window.

2.1 Definition of Signal and Background Training Samples

To successfully train a supervised Machine Learning (ML) classifier, we must provide it labeled datasets that accurately represent the pure signal and the pure background.

2.1.1 Signal Window and Background Extraction

The dominating background in our data sample is combinatorial. To train the classifier to recognize and reject this background, we extract a background training sample directly from the real data. However, we cannot simply use the entire dataset, as it contains our target signal events.

First, we must define a **signal region** (or signal window). Using the $B_s^0 \rightarrow \psi(2S)K_S^0$ Monte Carlo simulation, we analyze the invariant mass distribution of the pure signal. The signal window is defined as the shortest mass interval containing 99% of the simulated signal events (see figure 4).

Crucially, the corresponding mass region in the real data must be excluded during the training of the classifier. Including data from the expected signal region in the background training sample would bias the ML algorithm.

Instead, the background training sample is drawn from the **Upper Sideband (USB)**. The USB consists of events in the real data with a reconstructed invariant mass larger than the nominal B_s^0 mass (above the upper bound of the defined signal window). These events are entirely combinatorial background, making the USB an unbiased training sample.

2.1.2 Signal Training Sample and the Control Channel

For the signal training sample, we utilize the Monte Carlo simulation of the B_s^0 decay. However, relying purely on simulation introduces a significant challenge: simulations are not perfect due to constraints on CPU time for generating millions of events, as well as imperfect theoretical models and detector response approximations. Simulated kinematic distributions can deviate from real-world data. If we train our classifier naively, it might learn to distinguish "data vs. simulation" rather than "background vs. signal."

To mitigate this, two primary correction techniques are employed:

1. **Kinematic Weights:** The simulated samples are provided with pre-computed *kinematic weights*. These weights re-shape the simulated kinematic distributions to better align with data distributions, correcting mismodelings. Note that these weights are not
2. **Validation via Control Channel ($B^0 \rightarrow \psi(2S)K_S^0$):** Even after applying kinematic weights, some variables may still exhibit significant data-simulation discrepancies. Because we cannot validate our B_s^0 simulation against B_s^0 data (finding the data is the goal of the analysis), we employ a *control channel*. The decay $B^0 \rightarrow \psi(2S)K_S^0$ is abundant in our dataset and is kinematically similar to our signal decay. Therefore, any detector biases or simulation flaws will affect the B^0 simulation in a similar manner as the B_s^0 simulation.

To extract the pure data distributions for the B^0 control channel without background contamination, a statistical background-subtraction technique known as the **sPlot** method is used. This provides *sWeights* for the real data. Figure 5 compares this extracted data and the actual data. By comparing the sWeighted real data of the B^0 peak to the kinematically weighted B^0 simulation, we can evaluate the agreement of every potential input feature. Only variables that demonstrate data-simulation agreement in the control channel are passed to the classifier.

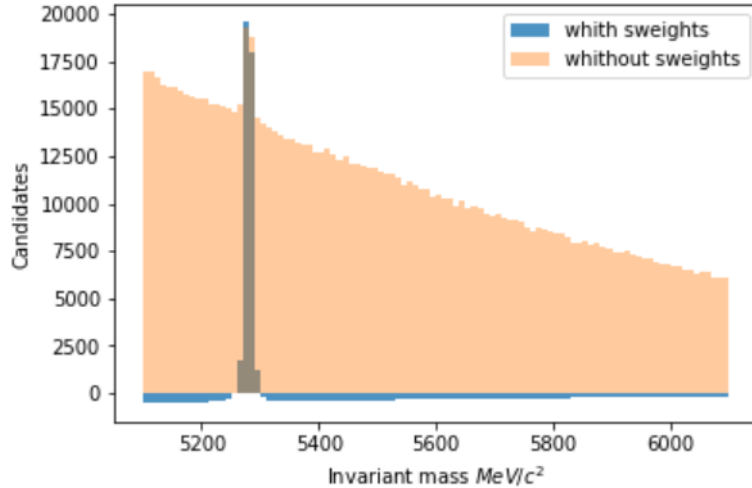


Figure 5: The plot shows a comparison of the invariant mass of B^0 meson with (blue) and without (orange) sWeights.

3 Feature Selection

Before training a multivariate classifier, it is critical to select a robust set of input variables (features). Feeding the classifier with poorly modeled or highly correlated features can lead to low performance, artificial biases, and an inability to properly extract the signal yield in the final analysis. The feature selection process in this study is governed by these requirements: data-simulation agreement, strong background-simulation discrimination, minimal correlation with the invariant mass and low correlation with other features.

3.1 Data-Simulation Agreement and Background Discrimination

As established, the signal training sample relies entirely on Monte Carlo simulation. If the simulation fails to accurately model a specific kinematic or geometric variable, the classifier might learn to distinguish between real data and simulation artifacts rather than true signal and background. This phenomenon leads to an overestimation of the classifier's performance during training and a low generalization.

To validate the features, we first compare the distributions of every available variable using the $B^0 \rightarrow \psi(2S)K_S^0$ control channel. Specifically, we compare the sWeighted real data (which isolates the pure B^0 distribution from the background) against the kinematically weighted B^0 MC simulation.

The agreement is quantified using the **Kolmogorov-Smirnov (KS) test**. The KS test computes the maximum distance between the empirical cumulative distribution functions of the two samples. Variables that exhibit a large KS distance, bigger than 0.5 (indicating poor modeling in the simulation) are rejected. (Figure 6)

$$\sup_n |F_n^1 - F_n^2| \quad (1)$$

On the selected features above we apply the same process for signal-background discrimination using kinematically weighted B^0 MC simulation as the signal and USB as the background. Variables that exhibit a small KS distance, smaller than 0.5, are rejected. (Figure 7)

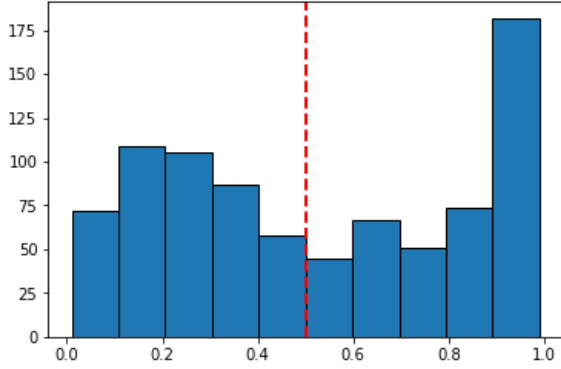


Figure 6: Figure showing a histogram of features binned according to their KS distance calculated for data-simulation agreement. We have discarded the features with KS distance above 0.5.

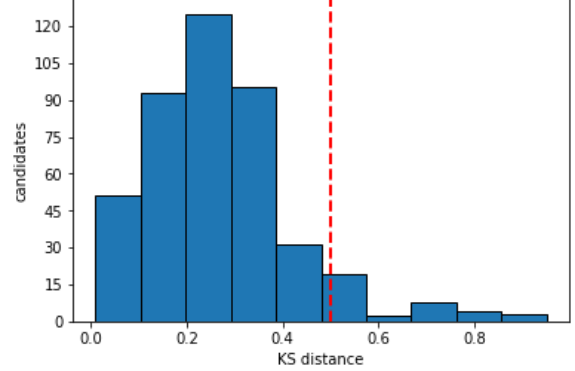


Figure 7: Figure showing a histogram of features binned according to their KS distance calculated for simulation-background agreement. We have ignored the features with KS distance below 0.5.

3.2 Correlation with Invariant Mass

The ultimate goal of this analysis is to determine the number of signal events by performing an unbinned maximum likelihood fit to the invariant mass distribution. To do this reliably, the background must maintain a smooth, predictable shape (typically an exponential decay) in the mass spectrum.

If a feature used to train the classifier is highly correlated with the candidate's invariant mass, applying a cut on the classifier's output will alter the shape of the invariant mass distribution. This effect, known as **mass sculpting**, can create artificial peaks in the combinatorial background that mimic a real signal, completely invalidating the final fit.

Therefore, we compute the correlation between every candidate feature and the invariant mass. Any variable showing a significant correlation (>0.5) with the mass is removed from the feature set.

3.3 Inter-Feature Correlation (Collinearity)

In addition if two kinematic variables are highly correlated with each other (collinear), they effectively provide the classifier with redundant information. Retaining redundant features increases the dimensionality of the model, and can make algorithms like Boosted Decision Trees (BDTs) more susceptible to overfitting.

Therefore we have computed the absolute linear correlation matrix for all remaining candidate features using the real data sample, as visible in Fig 9. A maximum correlation threshold is established at $|\rho_{max}| = 0.80$.

After applying these selection criteria, the remaining robust and uncorrelated variables, amounting to 20, are used to train the Machine Learning algorithm.

4 Training a Classification Algorithm

After having identified a set of discriminating features a Boosted Decision Tree (BDT), based on the XGBoost implementation [1], is used.

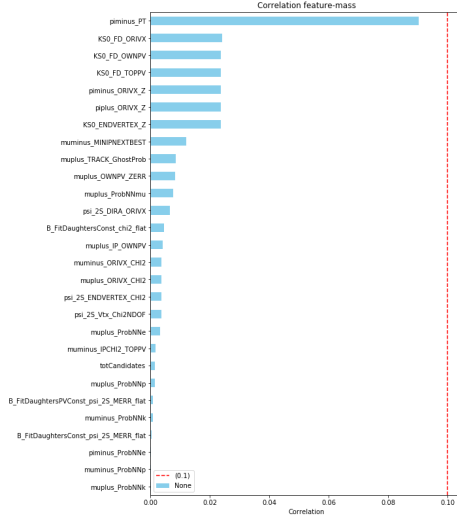


Figure 8: The figure shows the correlations among the different features after selecting those that exhibit good agreement between real and simulated data. The red vertical line indicates a correlation threshold of 0.1.

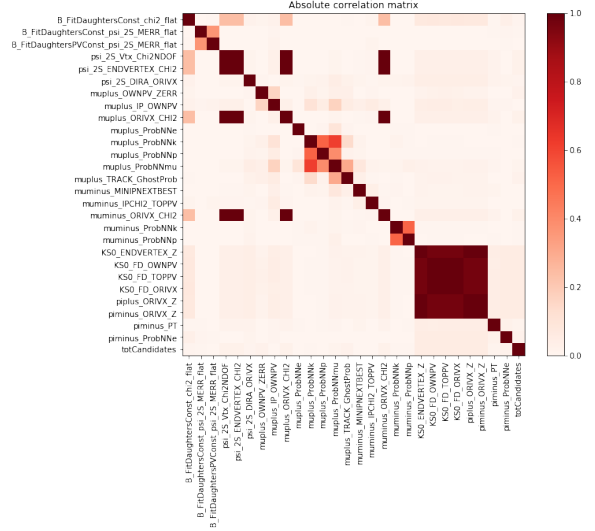


Figure 9: Absolute correlation matrix of the candidate features. The heatmap visually highlights pairs of highly correlated variables. Features showing a correlation magnitude greater than 0.80 with another variable were discarded to reduce redundancy.

4.1 Training Dataset and Sample Weights

The training dataset is constructed by combining two samples: the signal Monte Carlo simulation and USB of the B_s invariant mass distribution in data. A significant discrepancy in the number of events contained in the dataset exists between the two samples, as the background USB contains considerably more events than the signal simulation. This is a problem since we can obtain good accuracy simply because the most of the events come from the USB. To mitigate this effect, sample weights w_i are assigned to each event. Signal events carry kinematic weights w_{kin} to correct for known simulation mismodelling of the decay kinematics. Background events are assigned uniform weights such that the total weighted sum of the background sample equals that of the signal:

$$w_{\text{bkg}} = \frac{\sum_i w_{\text{kin}, \text{sig}}}{N_{\text{bkg}}}, \quad (2)$$

4.2 Cross-Validated Training

To avoid overtraining a k -fold cross-validation strategy is adopted using `StratifiedKFold` from `scikit-learn` [2], with $k = 5$ folds. We have paid attention to preserve the signal-to-background ratio for the different folds by shuffling the dataset properly. For each fold a BDT is trained on $\frac{k-1}{k}$ of the training data and evaluated on $\frac{1}{k}$ fraction. Each BDT is configured with 200 estimators, a maximum tree depth of 4, a learning rate of 0.05, and subsampling fractions of 0.8 for both events and features per tree.

4.3 Classifier Output and Performance Evaluation

The BDT output score $p \in [0, 1]$ represents the probability for an event to be signal-like. Fig. 10, shows the distribution of this score for signal and background events in the training sample. A clear separation between the two classes is observed, with signal events accumulating near $p = 1$ and background events near $p = 0$.

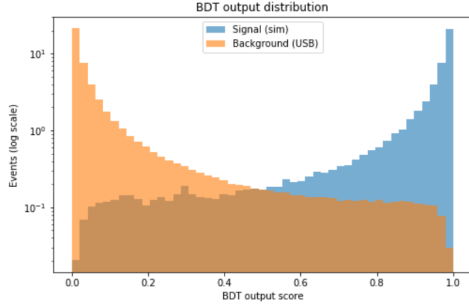


Figure 10: The figure show the distribution of the BDT output for the signal and background samples.

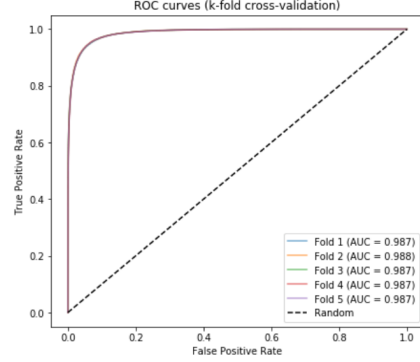


Figure 11: The figure shows the ROC curve obtained for the different folds, the dashed diagonal line represent the score expected for random selection.

The classifier performance is evaluated quantitatively through the Receiver Operating Characteristic (ROC) curve, which traces the true positive rate (signal efficiency) against the false positive rate (background retention) as a function of the threshold. The Area Under the Curve (AUC) is computed for each of the five folds, the result is visible in Fig. 11. The AUC obtain is very close to one for all the different folds with an average value of 0.985 ± 0.002 . Those very high values may be taken to suggest overfitting but since the number of different class is represented equally and the test is performed on the k folds on which the BDT is not trained, we can conclude that the classifier is not overfitting.

5 Classification Threshold Optimisation

With the trained classifier a threshold t on the BDT score must be chosen. The threshold is optimized by maximizing a figure of merit (FOM) that encodes the expected statistical sensitivity of the analysis. The FOM used in this analysis is defined as:

$$FOM = \frac{\varepsilon_{\text{sig}}(t)}{\frac{5}{2} + \sqrt{n_{\text{bkg}}(t)}}, \quad (3)$$

where $\varepsilon_{\text{sig}}(t)$ is the weighted signal efficiency at threshold t , defined as the fraction of signal simulation events with BDT score $p \geq t$, and $n_{\text{bkg}}(t)$ is the estimated number of background events after apply the cut. This expression is the Punzi figure of merit, Ref. [3], which is designed to optimise the sensitivity to a signal. The signal efficiency is computed on the signal simulation, taking into account the kinematic weights:

$$\varepsilon_{\text{sig}}(t) = \frac{\sum_{i: p_i \geq t} w_{\text{kin},i}}{\sum_i w_{\text{kin},i}}. \quad (4)$$

The number of background events in the signal window is estimated by fitting the background in the USB with an exponential function and integrating the obtained result in the region of the signal window $N_{\text{SR}}^{\text{raw}}$. The FOM defined in Eq. 3 is evaluated on two sets of threshold values: 100 uniformly spaced thresholds in the interval $[0, 0.9)$ and 100 additional thresholds in the interval $[0.9, 1)$, to obtain a finer scan in the region close to one. The FOM curve respect to the different cuts is visible in Fig. 12. The maximum is observed at $t_{\text{opt}} = 0.988$ and is $FOM_{\text{max}} = 0.049$.

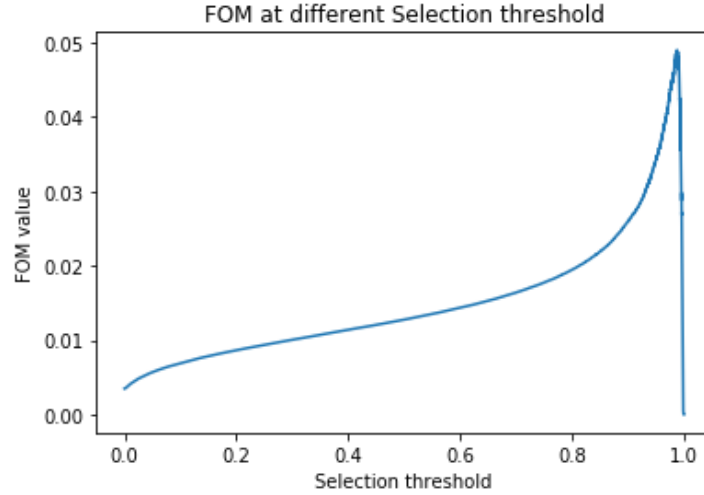


Figure 12: The figure show the plot obtained for the FOM for different threshold, a single peak is observed at $t_{\text{opt}} = 0.988$ and $FOM_{\text{max}} = 0.049$.

6 Fit signal peak

After training the BDT, the algorithm was applied to the invariant mass region in which the B^0 and B_s^0 peaks are expected in the real data sample. To improve the prediction all the models produced during the k -fold cross-validation procedure were used. For each event, the final classifier response was computed as the average of the probabilities predicted by the different folds. Events were then classified as signal candidates according to the optimal threshold t_{opt} obtained in Sec.5.

The events identified as signal-like in the two mass regions were used to perform a Gaussian fit to the invariant mass distributions. The resulting fits for both the contributions are shown in Fig. 13.

In addition, an exponential fit was performed on the USB invariant mass distribution in order to model the remaining combinatorial background contribution. The result of this fit is shown in Fig.14. The obtained results are an amplitude $A = 5.1 \pm 2.1 \cdot 10^5$ and $\lambda = 1.072 \pm 0.007 \text{ c}^2/\text{MeV}$.

The results obtained from the previous fit were used as starting point for the fit performed over the events selected by the BDT, visible in Fig 15. For the B^0 peak, the fit has provided a mean value of $\mu_{B^0} = 5279.84 \pm 0.12 \text{ MeV}/c^2$ with a standard deviation of $\sigma_{B^0} = 5.98 \pm 0.10 \text{ MeV}/c^2$. Similarly, for the B_s^0 signal region, the fit returns $\mu_{B_s^0} = 5366.2 \pm 1.2 \text{ MeV}/c^2$ with a corresponding width $\sigma_{B_s^0} = 4.58 \pm 0.89 \text{ MeV}/c^2$. It is possible to observe a good agreement with the value reported in the PDG, Ref, [4], for the invariant mass of the two bosons, $m_{B^0} = 5279.72 \pm 0.08 \text{ MeV}/c^2$ and $m_{B^0} =$

$5366.93 \pm 0.10 \text{ MeV}/c^2$. The overall agreement has been quantified computing the $\tilde{\chi}^2$ obtaining 1.63.

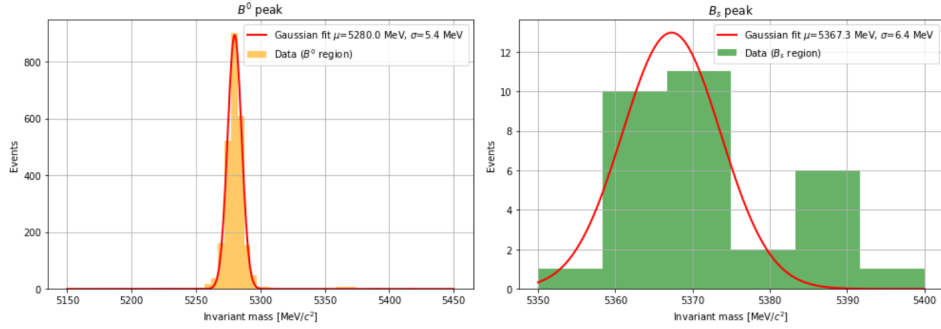


Figure 13: Gaussian fits performed on the invariant mass distributions after the BDT selection. The left panel shows the B^0 control-channel peak, while the right panel shows the B_s^0 signal candidate region.

The statistical significance is estimated using:

$$m = \frac{n_{\text{sig}}}{\sqrt{n_{\text{sig}} + n_{\text{bkg}}}}, \quad (5)$$

where n_{sig} and n_{bkg} denote the number of signal and background events within the signal region, respectively, obtained by integrating the corresponding analytical models over the signal interval, the obtained result is $m = 4.45 \sigma$.

7 Conclusion

In this laboratory exercise, we successfully implemented a multivariate analysis to isolate the rare $B_s^0 \rightarrow \psi(2S)K_S^0$ decay from a heavily polluted combinatorial background in LHCb data. By employing a Boosted Decision Tree (BDT) classifier, we were able to significantly enhance the signal-to-background ratio compared to traditional cut-based methods.

The robustness of the classifier was ensured through a rigorous feature selection process. By utilizing sWeighted data from the B^0 control channel and applying kinematic weights, we retained only features demonstrating excellent data-simulation agreement and strong background discrimination. Crucially, discarding variables highly correlated with the invariant mass successfully prevented mass sculpting, allowing for unbiased fits in the

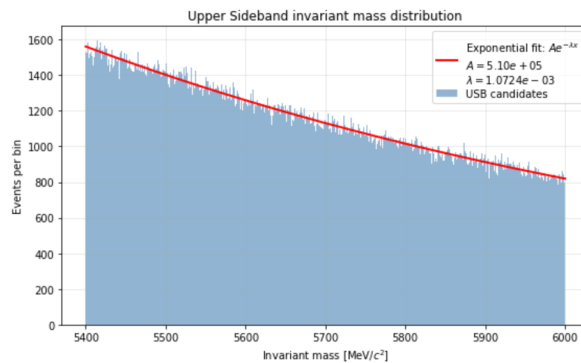


Figure 14: Exponential fit performed on the invariant mass distribution in the USB region.

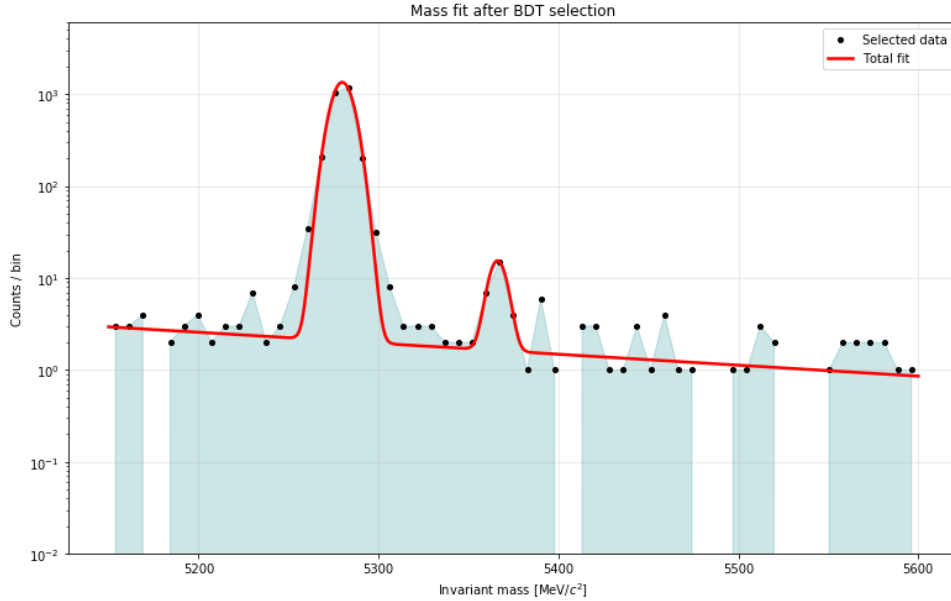


Figure 15: Combined fit to the invariant mass distribution in the region where the B^0 and B_s^0 signals are expected. The red curve represents the residual background contribution while the teal shaded area corresponds to the number of entries obtained from the selected events after the application of the BDT in the different bins. To increase the readability of the plot a logarithmic scale was used on the y-axis.

final analysis. The threshold optimization using the Punzi Figure of Merit yielded an optimal cut at $t_{\text{opt}} = 0.99$, highlighting the necessity of stringent selection criteria to filter this rare decay.

Following the BDT selection, the extracted invariant mass peaks demonstrated excellent agreement with the accepted experimental values from the Particle Data Group (PDG). The fitted mean mass for the B^0 control channel was found to be $\mu_{B^0} = 5279.98 \pm 0.06 \text{ MeV}/c^2$, which closely aligns with the PDG value of $5279.72 \pm 0.08 \text{ MeV}/c^2$. Similarly, the target B_s^0 signal peak was localized at $\mu_{B_s^0} = 5367.28 \pm 2.03 \text{ MeV}/c^2$, which is highly consistent with the known experimental mass of $5366.93 \pm 0.10 \text{ MeV}/c^2$.

Finally, the fits yielded a final statistical significance proxy of $m = 4.45$. This result represents strong evidence for the observation of the $B_s^0 \rightarrow \psi(2S)K_s^0$ decay within the heavily dominated background.

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